Gamma Determinants and Colored Stieltjes Moments

Closed normalization constants, finite Gaussian moment laws, and clustered polylogarithm identities

Research continuation prepared for Vladimir Reshetnikov's ProveIt programme

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Abstract

We resolve the incoming Gamma-basis normalization question at every dimension by reducing its coefficient determinant to Gustafson's classical multivariable Barnes integral. The resulting constants are explicit rational multiples of powers of π, with a two-step recurrence. A parameter deformation yields a further exact phenomenon: under an explicit positive n-variable Gamma measure, a normalized sum of trigamma values has precisely the first $n+1$ moments of a standard Gaussian. This holds at every positive Gamma shift and implies a Gaussian limit even when that shift varies with the dimension.

In a separate translation geometry, we solve the colored analogue of the incoming balanced Stieltjes-moment problem. A unique homogeneous combination of Lerch order derivatives cancels its first inverse-power tails. In the general convergence range, each positive-integer moment reduces to one negative-order Lerch value times a finite ordinary-zeta combination. Root-of-unity colors give explicit generalized Stieltjes identities; exceptional alternating moments evaluate to Dirichlet beta values. The color tending to one has an exact, normalization-dependent constant term, which we compute.

We also give a finite multiple-polylogarithm dictionary for Gamma spectral faces with two groups of repeated masses, and an explicit depth-two reduction for a printed Gaussian-color example. Classical input identities, new consequences relative to the inspected reports, numerical diagnostics, and remaining conjectures are kept distinct. In particular, the Gaussian S_6 and revised S_8 conjectures and the unrestricted unequal-tail prefix problem are not settled.

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1 Problems addressed and mathematical status

The current ProveIt manuscript develops polylogarithm identities together with harmonic-tail, Gamma, Hurwitz, and generalized Stieltjes calculi [1]. Recent incoming reports contain both explicit open questions and extensive proved developments. A continuation must therefore identify its exact starting point before claiming an advance.

The principal target here is Question 2 in the research section of *Ordered Boundary Identities and Gamma Kernels* [2]. That report constructs a Gamma integral basis for a fixed-gap harmonic recurrence and proves that its initial determinant equals $(-1)^{m(m-1)/2}K_m$, where K_m is a positive integral. It asks for closed evaluations of $K_2, K_3, \ldots$. We give one formula for every m, rather than a table of separately evaluated integrals.

The second target is Question 4 of *Balanced Stieltjes Moments* [3]: a colored or Lerch analogue of the homogeneous balancing and its polynomial moment reductions. The nontrivial color removes the order pole, so the correct generator has no separate $1/z$ subtraction. This changes the degree of the negative-order coordinate and makes the limit of color one singular. We prove the colored theorem on its full natural slit domain and give its exact connection with the uncolored normalization.

The repeated-mass theorem answers a two-cluster part of the spectral face question in [4]. It reduces arbitrary multiplicities to finitely many multiple polylogarithms. The number of lattice terms remains infinite: the finite dictionary describes the types and depths of polylogarithms in each term, and does not give a finite evaluation of the complete lattice sum.

1.1 A concise result ledger

Result	Exact advance	Status
Gamma normalization	Closed K_m for all m; explicit two-step recurrence	Incoming Question 2 resolved
Trigamma moments	After normalization, the first $n+1$ moments are exactly Gaussian for every positive shift	Proved consequence of a parameter identity
Colored balancing	Unique coefficients in a specified homogeneous ansatz; all integer moments and exceptional resonances	Colored counterpart resolved in that ansatz
Color-one limit	Explicit pole subtraction and a shift-dependent constant-term conversion	Proved normalization identity
Two mass groups	Finite multiple-polylogarithm formula at every pair of multiplicities	Two-cluster sector resolved
Depth-two parity	Exact double-shuffle and distribution certificate for one printed example	Classical parity made constructive
S_6, revised S_8	No exact certificate supplied here	Remain conjectural
Unequal-tail prefix	Initial basis normalization is evaluated; later determinant nonvanishing is a separate issue	Unrestricted problem remains open

1.2 Conventions and novelty

The Gamma function and its logarithmic derivative are denoted by Γ and ψ; $\psi^{(j)}$ always means argument differentiation. An order derivative of Hurwitz zeta or Lerch's transcendent is written

∂_s^j. The generalized Stieltjes normalization is

$$\zeta(1+z,a)=\frac{1}{z}+\sum_{j=0}^{\infty}\frac{(-1)^j\gamma_j(a)}{j!}z^j,\qquad \gamma_0(a)=-\psi(a). \tag{1.1}$$

All real-variable improper integrals asserted as ordinary integrals converge absolutely. A constant term is always taken in a specified parameter; no unspecified regularization is used.

Gustafson's integral, the Gamma reflection formula, the Lerch integral representation, Bernoulli and Euler special values, and multiple-polylogarithm Landen transformations are classical [5, 6, 8, 12, 11, 7]. Our normalization theorem is a new application to the specific incoming basis; the colored and clustered statements are explicit extensions of the inspected incoming formulas. The finite trigamma moment law is derived here from the complete parameter identity. A targeted literature search supports these comparisons but does not establish world-wide historical priority. No arithmetic independence, minimal period basis, or proof-assistant formalization is claimed.

2 An exact Gamma moment determinant

The incoming ordered-boundary report asks for closed evaluations of the positive constants K_m that normalize its Gamma integral basis [2, Section 8 and Section 9, question 2]. All of these constants can be evaluated. The decisive step is to remove the source's last moment by its already proved orthogonality relation and recognize the remaining determinant as Gustafson's symplectic Barnes integral. We first state the parameterized determinant with its full normalization. This is a reformulation of a classical integral, and it is credited as such; its application to the particular source constants and the moment identities developed below are the results of the present continuation.

For $n\ge 1$ and $\mathbf{a}=(a_1,\dots,a_{2n+2})$ satisfying $\Re a_\ell>0$, put

$$W_{\mathbf{a}}(x)=\prod_{\ell=1}^{2n+2}\Gamma(a_\ell+ix/2)\Gamma(a_\ell-ix/2),\qquad A=\sum_{\ell=1}^{2n+2}a_\ell. \tag{2.1}$$

Define the matrix and its determinant by

$$\mathsf{B}_n(\mathbf{a})_{r,k}=\int_0^\infty x^{2r-1}\sinh(k\pi x)W_{\mathbf{a}}(x)\,\mathrm{d}x,\qquad \mathcal{D}_n(\mathbf{a})=\det\mathsf{B}_n(\mathbf{a}),\quad 1\le r,k\le n. \tag{2.2}$$

The sinh columns are in increasing order. This convention is relevant when the result is applied to a source matrix with its columns reversed.

Theorem 2.1 (Parameterized Gamma moment determinant)**.** *For $\Re a_\ell>0$,*

$$\boxed{\mathcal{D}_n(\mathbf{a})=2^{2n^2}\pi^{n(n+1)}\frac{\prod_{\ell<j}\Gamma(a_\ell+a_j)}{\Gamma(A)}.} \tag{2.3}$$

Every entry is an absolutely convergent ordinary integral. The entries, their determinant, and every fixed mixed parameter derivative are holomorphic on this domain and may be differentiated under the integral. The determinant is nonzero throughout the domain and is positive when all parameters are positive real numbers.

Proof. Stirling's formula uniformly on compact parameter sets gives $W_{\mathbf{a}}(x)=O(x^Ce^{-(n+1)\pi x})$ for some finite C. The largest sinh factor grows as $e^{n\pi x}$. A fixed parameter derivative adds at most powers of $\log(2+x)$ and preserves an integrable exponential majorant. At zero all Gamma arguments remain in a compact subset of the right half-plane. These observations prove the asserted convergence and permit the uses of Fubini and parameter differentiation.

Write

$$\Delta(y_1,\dots,y_n)=\prod_{j<k}(y_k-y_j)$$

and introduce the integral

$$\begin{aligned}\mathcal{I}_n(\mathbf{a})=\int_{(0,\infty)^n}&\Delta(x_1^2,\dots,x_n^2)\Delta(\cosh\pi x_1,\dots,\cosh\pi x_n)\\&\times\prod_{j=1}^{n}x_j\sinh(\pi x_j)W_{\mathbf{a}}(x_j)\,\mathrm{d}x_1\cdots\mathrm{d}x_n.\end{aligned}\tag{2.4}$$

Expanding the two determinants in the continuous Cauchy–Binet identity and using Fubini gives

$$\mathcal{D}_n(\mathbf{a})=\frac{2^{n(n-1)/2}}{n!}\mathcal{I}_n(\mathbf{a}).\tag{2.5}$$

Indeed, the polynomial alternant is $\det(x_j^{2r-1})=\prod_j x_j\,\Delta(x_1^2,\dots,x_n^2)$, while

$$\det(\sinh(k\pi x_j))_{j,k=1}^{n}=2^{n(n-1)/2}\prod_j\sinh(\pi x_j)\Delta(\cosh\pi x_1,\dots,\cosh\pi x_n).$$

The second identity follows from $\sinh(ku)=\sinh(u)U_{k-1}(\cosh u)$ and the leading coefficient 2^{k-1} of the Chebyshev polynomial U_{k-1}.

For precision, the classical integral used next is Gustafson's symplectic Barnes integral [5], in the normalization reproduced in [6, equation (2)]:

$$\begin{aligned}&\int_{\mathbb{R}^n}\frac{\displaystyle\prod_{j=1}^{n}\prod_{\ell=1}^{2n+2}\Gamma(a_\ell+iu_j)\Gamma(a_\ell-iu_j)}{\displaystyle\prod_j\Gamma(2iu_j)\Gamma(-2iu_j)\prod_{j<k}\Gamma(iu_j+iu_k)\Gamma(iu_j-iu_k)\Gamma(-iu_j+iu_k)\Gamma(-iu_j-iu_k)}\,\mathrm{d}\mathbf{u}\\&\qquad=(2\pi)^n2^nn!\frac{\prod_{\ell<j}\Gamma(a_\ell+a_j)}{\Gamma(A)}.\end{aligned}\tag{2.6}$$

The right-half-plane hypotheses place all numerator poles on the appropriate sides of the imaginary integration axes in its Barnes form. The integrand is even in each u_j, so the positive-orthant integral equals $(2\pi)^n n!$ times the Gamma quotient.

Set $x_j=2u_j$. Reflection yields the single-variable factor

$$\frac{1}{\Gamma(ix_j)\Gamma(-ix_j)}=\frac{x_j\sinh(\pi x_j)}{\pi},\tag{2.7}$$

and each four-factor interaction contributes

$$\begin{aligned}&\left\{\Gamma\left(\tfrac{i}{2}(x_j+x_k)\right)\Gamma\left(-\tfrac{i}{2}(x_j+x_k)\right)\Gamma\left(\tfrac{i}{2}(x_k-x_j)\right)\Gamma\left(-\tfrac{i}{2}(x_k-x_j)\right)\right\}^{-1}\\&\qquad=\frac{(x_k^2-x_j^2)(\cosh\pi x_k-\cosh\pi x_j)}{8\pi^2}.\end{aligned}\tag{2.8}$$

The factor 2^{-n} from $\mathrm{d}\mathbf{u}$ must also be retained. Consequently (2.6) gives

$$\mathcal{I}_n(\mathbf{a})=2^{2n+3n(n-1)/2}\pi^{n(n+1)}n!\frac{\prod_{\ell<j}\Gamma(a_\ell+a_j)}{\Gamma(A)}.\tag{2.9}$$

Combining this with (2.5) proves (2.3), since the two powers of 2 add to $2n^2$. The Gamma quotient is nonzero in the stated domain. For positive parameters the two Vandermonde factors in (2.4) have the same sign, and the integrand is strictly positive away from collisions. This proves the positivity assertion as well. □

The case $n=0$ may be included with the empty determinant and empty integral both equal to one. Its Gamma quotient is then $\Gamma(a_1+a_2)/\Gamma(a_1+a_2)=1$.

3 Closed evaluation of every Gamma-basis normalization

We recall exactly the source data needed for the application. For an integer $m \geq 1$, set

$$W_m(x) = |\Gamma(\tfrac{1}{4} + ix/2)|^{4m}, \qquad \phi_r(x) = \Im(x + i/2)^{2r}, \tag{3.1}$$

$$P_j(x) = \sinh(\pi(m-j)x) \quad (1 \leq j < m), \qquad P_m(x) = \tanh(\pi x), \tag{3.2}$$

$$t_{r,j} = \frac{4}{(2r)!} \int_0^\infty \phi_r(x) P_j(x) W_m(x)\, \mathrm{d}x, \qquad T_m = (t_{r,j})_{r,j=1}^m. \tag{3.3}$$

The incoming report proves the exact top moments

$$t_{m,j} = 0 \quad (j < m), \qquad t_{m,m} = (-1)^{m+1} \frac{2^{2m+2}\pi^m}{(2m)!}, \tag{3.4}$$

by Gamma recurrence and a contour shift crossing precisely two tanh poles. It also proves

$$\det T_m = (-1)^{m(m-1)/2} K_m, \qquad K_m > 0. \tag{3.5}$$

Equations (3.4) and (3.5) are inherited results, specifically the source theorem `gb:basis`, corollary `gb:moment-identities`, and proposition `gb:normalization` [2]. The following evaluation answers the open question about the remaining positive constant.

Theorem 3.1 (All Gamma normalization constants)**.** *For every integer $m \geq 1$,*

$$\boxed{K_m = \frac{2^{2m^2+2}(m-1)!\,\pi^{m(4m-1)/2}}{\Gamma(m/2)\prod_{r=1}^m (2r)!}.} \tag{3.6}$$

In particular,

$$K_1 = 8\pi, \qquad K_2 = \frac{64}{3}\pi^7, \qquad K_3 = \frac{16384}{135}\pi^{16}. \tag{3.7}$$

The constants satisfy the two-step recurrence

$$\boxed{\frac{K_{m+2}}{K_m} = \frac{2^{8m+9}(m+1)\pi^{8m+7}}{(2m+2)!(2m+4)!}.} \tag{3.8}$$

Each K_m is a positive rational multiple of $\pi^{2m^2-\lceil m/2 \rceil}$.

Proof. Expand $\det T_m$ along the last row, using (3.4). Since $\phi_r(x) = rx^{2r-1}$ plus lower odd powers, triangular row elimination in the remaining $(m-1) \times (m-1)$ minor extracts

$$\prod_{r=1}^{m-1} \frac{4r}{(2r)!}.$$

Reversing the source sinh-column order $m-1, \ldots, 1$ to the increasing order $1, \ldots, m-1$ contributes the sign $(-1)^{(m-1)(m-2)/2}$. The remaining determinant is exactly $\mathcal{D}_{m-1}(1/4, \ldots, 1/4)$, so

$$\begin{aligned} \det T_m = (-1)^{m+1+(m-1)(m-2)/2} \frac{2^{2m+2}\pi^m}{(2m)!} \left(\prod_{r=1}^{m-1} \frac{4r}{(2r)!} \right) \\ \times \mathcal{D}_{m-1}(1/4, \ldots, 1/4). \end{aligned} \tag{3.9}$$

The sign equals $(-1)^{m(m-1)/2}$: the exponents differ by two. Theorem 2.1, with $2m$ quarter parameters, gives

$$\mathcal{D}_{m-1}(1/4, \ldots, 1/4) = 2^{2(m-1)^2} \pi^{m(m-1)} \frac{\Gamma(1/2)^{m(2m-1)}}{\Gamma(m/2)}. \tag{3.10}$$

Substitute $\Gamma(1/2) = \sqrt{\pi}$, compare with (3.5), and collect the factorials and powers. This yields (3.6). At $m = 1$ the minor is empty, so the same computation remains valid. Taking the quotient of the formulas at $m+2$ and m, and using $\Gamma(m/2+1) = (m/2)\Gamma(m/2)$, proves (3.8). The final assertion follows from the integer and half-integer Gamma evaluations. □

The result makes the inherited exact Gamma determinant representation fully explicit. For example, if $F_{b,m}$ is the rational fundamental matrix of the source recurrence, its already proved identity becomes

$$\det F_{b,m} = (-1)^{m(m-1)/2} \frac{\Gamma(m/2) \prod_{r=1}^{m} (2r)!}{2^{2m^2+2}(m-1)! \pi^{m(4m-1)/2}} \det(t_{b+i,j})_{i,j=1}^{m}. \tag{3.11}$$

This evaluates a previously unspecified normalization. It preserves the source's distinction between its eventual fixed-gap nonvanishing theorem and its still open all-pairs reconstruction conjecture.

4 Paired Gamma deformations and polygamma integrals

The parameter freedom in Theorem 2.1 is useful beyond the single quarter specialization. For $n \geq 1$, $\sigma > 0$, and $m = n + 1$, take

$$a_{2r-1} = \sigma + u_r, \qquad a_{2r} = \sigma - u_r, \qquad 1 \leq r \leq m, \qquad |\Re u_r| < \sigma. \tag{4.1}$$

Let $\mathcal{D}_{n,\sigma}(\mathbf{u})$ denote the corresponding determinant. Every pair has the same sum 2σ, so the denominator in (2.3) remains fixed.

Theorem 4.1 (Complete paired deformation)**.** *On the domain* (4.1)*,*

$$\boxed{\frac{\mathcal{D}_{n,\sigma}(\mathbf{u})}{\mathcal{D}_{n,\sigma}(\mathbf{0})} = \prod_{r<s} \frac{\prod_{\varepsilon,\eta\in\{-1,1\}} \Gamma(2\sigma + \varepsilon u_r + \eta u_s)}{\Gamma(2\sigma)^4}.} \tag{4.2}$$

At the quarter parameter this is elementary:

$$\boxed{\frac{\mathcal{D}_{n,1/4}(\mathbf{u})}{\mathcal{D}_{n,1/4}(\mathbf{0})} = \prod_{r<s} \sec(\pi(u_r + u_s)) \sec(\pi(u_r - u_s)).} \tag{4.3}$$

If only one pair is deformed, the two respective ratios are

$$\left\{ \frac{\Gamma(2\sigma + u)\Gamma(2\sigma - u)}{\Gamma(2\sigma)^2} \right\}^{2n}, \qquad \sec^{2n}(\pi u). \tag{4.4}$$

Proof. The m within-pair Gamma factors are $\Gamma(2\sigma)$ and are unchanged. Each pair of distinct labels contributes the four factors shown in (4.2); all remaining constants cancel. At $\sigma = 1/4$, apply $\Gamma(1/2 + z)\Gamma(1/2 - z) = \pi/\cos(\pi z)$ first to $z = u_r + u_s$ and then to $z = u_r - u_s$. The single-pair specialization has n identical pair interactions and gives (4.4). □

4.1 A positive density and its exact normalization

Define

$$\begin{aligned} \rho_{n,\sigma}(\mathbf{x}) = \Delta(x_1^2, \dots, x_n^2)\Delta(\cosh \pi x_1, \dots, \cosh \pi x_n) \\ \times \prod_{j=1}^{n} x_j \sinh(\pi x_j) |\Gamma(\sigma + i x_j/2)|^{4(n+1)}, \qquad \mathbf{x} \in (0,\infty)^n. \end{aligned} \tag{4.5}$$

The density is symmetric and nonnegative, and is positive off the collision hyperplanes. Its normalizing integral is

$$\mathcal{Z}_{n,\sigma} = 2^{n(3n+1)/2} \pi^{n(n+1)} n! \frac{\Gamma(2\sigma)^{(n+1)(2n+1)}}{\Gamma(2(n+1)\sigma)}. \tag{4.6}$$

This follows from (2.9). Write $\mathbb{E}_{n,\sigma}$ for expectation with respect to $\rho_{n,\sigma}/\mathcal{Z}_{n,\sigma}$, and define the real linear statistics

$$S_q = \sum_{j=1}^{n} \Re\psi^{(q)}(\sigma + ix_j/2), \qquad q = 1, 3, 5, \dots. \tag{4.7}$$

Theorem 4.2 (All-order polygamma insertion identity)**.** *As an identity of holomorphic germs at* $\mathbf{u} = \mathbf{0}$*,*

$$\begin{aligned} &\mathbb{E}_{n,\sigma} \exp\left(4 \sum_{r=1}^{n+1} \sum_{k\geq 1} \frac{u_r^{2k}}{(2k)!} S_{2k-1}\right) \\ &\qquad = \prod_{r<s} \frac{\prod_{\varepsilon,\eta\in\{-1,1\}} \Gamma(2\sigma + \varepsilon u_r + \eta u_s)}{\Gamma(2\sigma)^4}. \end{aligned} \tag{4.8}$$

In particular, every finite coefficient gives an ordinary absolutely convergent integral identity in polygamma functions.

Proof. Apply continuous Cauchy–Binet to the deformed determinant and divide by its value at zero. The interaction Vandermondes are unchanged, so the quotient is the expectation of the product of Gamma-factor ratios. For one pair and $z = \sigma + ix/2$, the logarithm of that ratio is

$$\begin{aligned} &\log\Gamma(z+u) + \log\Gamma(z-u) - 2\log\Gamma(z) \\ &\quad + \log\Gamma(\overline{z}+u) + \log\Gamma(\overline{z}-u) - 2\log\Gamma(\overline{z}) \\ &\qquad = 4\sum_{k\geq 1} \frac{u^{2k}}{(2k)!} \Re\psi^{(2k-1)}(z). \end{aligned}$$

The logarithmic Gamma branches are holomorphic on the right half-plane. Their Taylor expansions converge uniformly for $|u|$ in a fixed sufficiently small disk, with the integral dominated as in Theorem 2.1. Alternatively every fixed coefficient can be obtained directly by differentiating the finite Gamma product under its normally convergent integral. Multiplication over spatial variables and pair labels gives the left side of (4.8); Theorem 4.1 evaluates it. □

For instance, the single-pair coefficients at degrees two and four are

$$\mathbb{E}_{n,\sigma} S_1 = n\psi'(2\sigma), \tag{4.9}$$

$$\mathbb{E}_{n,\sigma}(S_3 + 12S_1^2) = n\psi'''(2\sigma) + 12n^2\psi'(2\sigma)^2. \tag{4.10}$$

The complete paired deformation separates the two terms on the left of the second identity and yields a stronger moment law.

5 Exact finite Gaussian moments

Theorem 5.1 (Gaussian moment matching through degree $n+1$)**.** *Let* $n \geq 1$*,* $\sigma > 0$*, and set*

$$\mu_\sigma = \psi'(2\sigma) > 0, \qquad v_\sigma = \psi'''(2\sigma) > 0, \qquad S = S_1. \tag{5.1}$$

For every integer $0 \leq k \leq n+1$*,*

$$\boxed{\mathbb{E}_{n,\sigma} S^k = \sum_{j=0}^{\lfloor k/2 \rfloor} \frac{k!}{2^j j!(k-2j)!} (n\mu_\sigma)^{k-2j} (v_\sigma/4)^j.} \tag{5.2}$$

Equivalently, the statistic

$$X_{n,\sigma} = \frac{2S - 2n\mu_\sigma}{\sqrt{v_\sigma}} \tag{5.3}$$

has exactly the standard Gaussian moments through degree $n+1$:

$$\mathbb{E}_{n,\sigma} X_{n,\sigma}^k = \begin{cases} 0, & k \text{ odd}, \\ (k-1)!!, & k \text{ even}, \end{cases} \qquad 0 \le k \le n+1, \tag{5.4}$$

where the moment at degree zero is one.

Proof. Choose k distinct pair labels in (4.8) and extract the coefficient of $u_1^2 \cdots u_k^2$. Put $w_r = u_r^2$ and compute in the quotient in which $w_r^2 = 0$. On the left, all terms with u_r^4 or higher disappear, so the required coefficient is $\mathbb{E}_{n,\sigma}(2S)^k$.

For one pair interaction, Taylor expansion of its logarithm gives

$$\sum_{\varepsilon,\eta\in\{-1,1\}} \log\Gamma(2\sigma + \varepsilon u_r + \eta u_s) - 4\log\Gamma(2\sigma)$$
$$\equiv 2\mu_\sigma(w_r + w_s) + v_\sigma w_r w_s \pmod{w_r^2, w_s^2}.$$

The sum is even in each pair variable. Each of the $n+1$ labels has exactly n partners, and no logarithmic summand depends on three distinct labels. Thus the logarithm of the complete right side reduces to

$$2n\mu_\sigma \sum_{r=1}^{k} w_r + v_\sigma \sum_{r<s\le k} w_r w_s.$$

To obtain $w_1 \cdots w_k$ from its exponential, partition the k labels into singleton and pair blocks. The number of partitions with j pairs is $k!/(2^j j!(k-2j)!)$. Such a partition contributes $(2n\mu_\sigma)^{k-2j} v_\sigma^j$. Therefore

$$\mathbb{E}_{n,\sigma}(2S)^k = \sum_{j=0}^{\lfloor k/2 \rfloor} \frac{k!}{2^j j!(k-2j)!} (2n\mu_\sigma)^{k-2j} v_\sigma^j.$$

Division by 2^k proves (5.2). These are the moments of a Gaussian variable with mean $n\mu_\sigma$ and variance $v_\sigma/4$. Centering and scaling proves (5.4). The restriction $k \le n+1$ is exactly the number of distinct pair parameters available in the proof. □

Corollary 5.2 (Separated low-order polygamma integrals)**.** *For all $n \ge 1$ and $\sigma > 0$,*

$$\mathrm{Var}_{n,\sigma}(S_1) = \frac{1}{4}\psi'''(2\sigma), \qquad \mathbb{E}_{n,\sigma} S_3 = (n-3)\psi'''(2\sigma). \tag{5.5}$$

At the quarter parameter,

$$\mathbb{E}_{n,1/4} S_1 = \frac{n\pi^2}{2}, \qquad \mathrm{Var}_{n,1/4}(S_1) = \frac{\pi^4}{4}, \qquad \mathbb{E}_{n,1/4} S_3 = (n-3)\pi^4. \tag{5.6}$$

Proof. The degree-two instance of (5.2) gives the variance. Insert the resulting second moment into (4.10) to obtain the expectation of S_3. Finally, $\psi'(1/2) = \pi^2/2$ and $\psi'''(1/2) = \pi^4$ give the quarter values [10]. □

5.1 Explicit one-dimensional identities

At $n = 1$ and $\sigma = 1/4$, the normalizer is $4\pi^5$. To display several resulting evaluations compactly, write

$$\omega(x) = x\sinh(\pi x)|\Gamma(\tfrac{1}{4} + ix/2)|^8, \qquad z_x = \tfrac{1}{4} + ix/2.$$

Every integral below is an ordinary absolutely convergent integral:

$$\int_0^\infty \omega(x)\,\mathrm{d}x = 4\pi^5, \tag{5.7}$$

$$\int_0^\infty \omega(x)\Re\psi'(z_x)\,\mathrm{d}x = 2\pi^7, \tag{5.8}$$

$$\int_0^\infty \omega(x)[\Re\psi'(z_x)]^2\,\mathrm{d}x = 2\pi^9, \tag{5.9}$$

$$\int_0^\infty \omega(x)\Re\psi'''(z_x)\,\mathrm{d}x = -8\pi^9. \tag{5.10}$$

The negative sign in (5.10) is compatible with an ordinary real integral: the real part of the complex-argument polygamma function changes sign.

Unpaired parameter derivatives supply digamma identities as well. Differentiating the $n = 1$ version of (2.3) once in one parameter and twice in two distinct parameters, then setting all four parameters equal to $1/4$, gives

$$\int_0^\infty \omega(x)\Re\psi(z_x)\,\mathrm{d}x = -4\pi^5(\gamma + 3\log 2), \tag{5.11}$$

$$\int_0^\infty \omega(x)[\Re\psi(z_x)]^2\,\mathrm{d}x = 4\pi^5\left((\gamma + 3\log 2)^2 + \frac{\pi^2}{12}\right). \tag{5.12}$$

For completeness, the first logarithmic parameter derivative of the Gamma quotient is $3\psi(1/2) - \psi(1)$, while the mixed logarithmic derivative in distinct labels is $\psi'(1/2) - \psi'(1)$. The integral insertions are respectively $2\Re\psi(z_x)$ and $4[\Re\psi(z_x)]^2$. These factors give exactly (5.11)–(5.12).

5.2 A distributional consequence of the exact identities

The Gaussian matching is asserted only through the stated finite degree. For fixed n and σ the statistic is bounded, since

$$|\Re\psi'(\sigma + iy)| \le \sum_{r=0}^{\infty} \frac{1}{(r+\sigma)^2} = \psi'(\sigma).$$

Thus it has a compactly supported distribution. Its moment identities nevertheless imply a Gaussian limit as the dimension increases.

Corollary 5.3 (Gaussian limit for arbitrary positive parameter sequences)**.** *For any sequence $\sigma_n > 0$, the random variables X_{n,σ_n} of (5.3) converge in distribution to a standard normal random variable as $n \to \infty$.*

Proof. For every fixed integer k, the kth moment is exactly the standard Gaussian moment once $n \ge k - 1$, independently of σ_n. The second moment equals one for every $n \ge 1$, giving tightness. For each fixed k, a higher even moment is uniformly bounded along the tail of the sequence, so the kth powers are uniformly integrable. Every subsequential weak limit therefore has all standard Gaussian moments. The normal distribution is moment determinate; for example its even moments satisfy Carleman's condition. Consequently every subsequential limit is standard normal, proving the assertion. □

This consequence uses exact moment identities at each sufficiently large finite dimension. No uniform large-dimension estimate on Gamma functions or on the parameter sequence is required.

6 Colored balanced kernels and their Mellin transform

Let

$$\Omega = \mathbb{C} \setminus [1, \infty), \qquad a > 0, \qquad x > 0, \qquad \omega \in \Omega.$$

The branch of $\Phi(\omega, s, a)$ is determined by its defining series for $|\omega| < 1$ and continuation to Ω. Equivalently, for $\Re s > 0$ it is given by the classical absolutely convergent integral [12, 25.14.5]

$$\Phi(\omega, s, a) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1} e^{-at}}{1 - \omega e^{-t}} \, \mathrm{d}t. \tag{6.1}$$

Subtracting Taylor coefficients of its smooth density at zero extends it to an entire function of s for each $\omega \in \Omega$.

Definition 6.1 (Colored balanced derivative)**.** For an integer $q \geq 0$, put

$$F_\omega(z; x, a) = x^z \Phi(\omega, 1 + z, x + a), \tag{6.2}$$

$$\begin{aligned} C_q^\omega(x, a) &= \partial_z^q F_\omega(z; x, a)|_{z=0} \\ &= \sum_{j=0}^{q} \binom{q}{j} \log^{q-j} x \, \partial_s^j \Phi(\omega, 1, x + a). \end{aligned} \tag{6.3}$$

Order derivatives are evaluated after analytic continuation in ω.

There is no order pole at $s = 1$ when $\omega \neq 1$ in this domain. Thus adding the uncolored term $\log^{q+1} x/(q + 1)$ to (6.3) would destroy decay rather than improve it.

6.1 Tail cancellation and uniqueness

Write

$$A_k(\omega, a) = \Phi(\omega, -k, a) = (\omega \partial_\omega + a)^k \frac{1}{1 - \omega}. \tag{6.4}$$

These are rational functions of ω and polynomials of degree k in a. Their leading coefficient in a is $1/(1 - \omega)$.

Lemma 6.2 (Uniform colored expansion)**.** *On compact subsets of $\Omega \times (0, \infty)$, and locally uniformly for z near zero,*

$$F_\omega(z; x, a) = \sum_{k=0}^{K} \frac{(-1)^k A_k(\omega, a)}{k!} (1 + z)_k x^{-k-1} + O(x^{-K-2}). \tag{6.5}$$

Consequently

$$C_q^\omega(x, a) = \sum_{k=q}^{K} \frac{(-1)^k A_k(\omega, a)}{k!} \, \partial_z^q (1 + z)_k|_{z=0} \, x^{-k-1} + O(x^{-K-2}), \tag{6.6}$$

and the leading coefficient is $(-1)^q A_q(\omega, a) x^{-q-1}$. At zero, $C_q^\omega(x, a) = O(1 + |\log x|^q)$.

Proof. The density $e^{-at}/(1 - \omega e^{-t})$ is analytic at $t = 0$, with Taylor coefficients $(-1)^k A_k(\omega, a)/k!$. Taylor's formula on a small fixed interval and exponential decay outside that interval give a Laplace expansion in (6.1). The kth term integrates to $\Gamma(1 + z + k) x^{-1-z-k}$; division by $\Gamma(1 + z)$ and multiplication by x^z give (6.5). On the stated compact sets, the denominator stays uniformly away from zero along $t \geq 0$, so the remainders are locally uniform. Cauchy differentiation in a slightly larger z disc preserves the inverse-power order of the remainder. Since $(1 + z)_k$ has degree k, derivatives of order $q > k$ vanish, and its leading coefficient gives $\partial_z^q (1 + z)_q = q!$. The assertion at zero follows directly from (6.3). □

Theorem 6.3 (Unique homogeneous balancing)**.** *Fix $\omega \in \Omega$ and $q \geq 0$. Among combinations*

$$D(x,a) = \sum_{j=0}^{q} d_j \log^{q-j} x \, \partial_s^j \Phi(\omega, 1, x+a), \qquad d_0 = 1,$$

with coefficients independent of x, a, the condition $D(x,a) = O(x^{-q-1})$ for every a in a nonempty open interval contained in $(0,\infty)$ forces $d_j = \binom{q}{j}$. These coefficients do satisfy that condition.

Proof. Put $d(t) = \sum_{j=0}^{q} d_j t^j$ and $L = \log x$. The coefficient arising from the kth density term is a polynomial in L. Its component with L^{q-r} contains $d^{(r)}(-1)$ times a nonzero scalar and $\partial_s^r (s)_k|_{s=1}$. For $k = r < q$, the latter derivative is $r!$. The polynomial $A_r(\omega, a)$ cannot vanish on the whole prescribed interval. Successively removing $x^{-1}, \ldots, x^{-q}$ therefore gives $d^{(r)}(-1) = 0$ for $r = 0, \ldots, q-1$. Since $\deg d \leq q$ and $d(0) = 1$, $d(t) = (1+t)^q$. The converse is Lemma 6.2. This proves uniqueness only in the displayed ansatz, not among arbitrary x-dependent combinations. □

Remark 6.4 (The interval in the shift is essential)*.* The uniqueness statement can fail if imposed only at one shift. Set

$$a_* = \frac{1+\sqrt{2}}{2}, \qquad \omega_* = -3 - 2\sqrt{2} \in \Omega.$$

The rational second moment factors as

$$A_2(\omega, a_*) = \frac{(3-2\sqrt{2})(\omega + 3 + 2\sqrt{2})^2}{4(1-\omega)^3}.$$

Consequently $A_2(\omega_*, a_*) = 0$ and, by $A_3 = (\omega\partial_\omega + a_*)A_2$, $A_3(\omega_*, a_*) = 0$ as well. The uniform tail formula therefore gives

$$C_2^{\omega_*}(x, a_*) = O(x^{-5}), \qquad C_3^{\omega_*}(x, a_*) = O(x^{-5}).$$

Hence the different homogeneous combination

$$D(x) = 2C_3^{\omega_*}(x, a_*) - \log x \, C_2^{\omega_*}(x, a_*)$$

also satisfies $D(x) = O(x^{-4})$. In the displayed uniqueness ansatz its coefficient vector is $\boldsymbol{d} = (1, 4, 5, 2)$ rather than $(1, 3, 3, 1)$, so it has the same normalization $d_0 = 1$. The fourth moment is nonzero, $A_4(\omega_*, a_*) = (2-\sqrt{2})/64$, and the modified kernel is not identically the canonical one. Requiring decay for a nonempty open interval of positive shifts excludes this pointwise degeneracy exactly as the proof of the theorem requires.

6.2 The exact Mellin identity

Define the generalized harmonic polynomial $Q_q(u)$ by

$$\frac{\Gamma(u+z)}{\Gamma(u)\Gamma(1+z)} = \sum_{q=0}^{\infty} Q_q(u) \frac{z^q}{q!}. \tag{6.7}$$

With $H_{u-1}^{(1)} = \psi(u) + \gamma$ and $H_{u-1}^{(r)} = \zeta(r) - \zeta(r,u)$ for $r \geq 2$, this is

$$\exp\left(\sum_{r=1}^{\infty} \frac{(-1)^{r-1}}{r} H_{u-1}^{(r)} z^r \right).$$

In particular, $Q_0 = 1$, $Q_1 = H^{(1)}$, and $Q_2 = (H^{(1)})^2 - H^{(2)}$.

Theorem 6.5 (Colored balanced Mellin formula)**.** *For $q \geq 0$ and $0 < \Re u < q + 1$,*

$$\boxed{\int_0^\infty x^{u-1} C_q^\omega(x, a)\, \mathrm{d}x = \frac{\pi}{\sin \pi u}\, Q_q(u)\Phi(\omega, 1-u, a).} \tag{6.8}$$

Every apparent singularity at an interior integer is removable. The integral is absolutely convergent and locally uniform in u, ω, a on the stated domains.

Proof. Initially take $0 < \Re u < 1$ and z sufficiently close to zero. Insert (6.1) into (6.2). Absolute Fubini and the Gamma integral give

$$\begin{aligned}\int_0^\infty x^{u-1} F_\omega(z; x, a)\, \mathrm{d}x &= \frac{\Gamma(u+z)}{\Gamma(1+z)} \int_0^\infty \frac{t^{-u} e^{-at}}{1-\omega e^{-t}}\, \mathrm{d}t \\ &= \frac{\Gamma(u+z)\Gamma(1-u)}{\Gamma(1+z)} \Phi(\omega, 1-u, a).\end{aligned}$$

The behavior at zero and infinity in Lemma 6.2 justifies locally normal differentiation in z. Euler reflection then proves (6.8) on the initial strip. The differentiated kernel decays as x^{-q-1}, so its integral is holomorphic on the larger strip $0 < \Re u < q + 1$. Analytic continuation establishes the formula there and makes the interior integer poles removable. This does not differentiate a divergent integral at an integer. □

7 Integer colored moments, Stieltjes differences, and primitives

For $q \geq m \geq 1$, let

$$L_{m,q} := (-1)^q q! \sum_{\ell=m}^{\infty} \frac{\binom{\ell-1}{m-1}}{\ell^{q+1}} \tag{7.1}$$

$$= (-1)^{m+q+1} q! \sum_{j=0}^{m-1} (-1)^j e_j(1, \tfrac{1}{2}, \ldots, \tfrac{1}{m-1}) \zeta(q+1-j). \tag{7.2}$$

The first representation is convergent and gives its sign. The second is a finite zeta expression, with the empty-list convention $e_0 = 1$.

Theorem 7.1 (All ordinary positive-integer moments)**.** *For $q \geq m \geq 1$,*

$$\boxed{\int_0^\infty x^{m-1} C_q^\omega(x, a)\, \mathrm{d}x = L_{m,q} \Phi(\omega, 1-m, a).} \tag{7.3}$$

Thus the complete shift dependence is a polynomial of degree $m-1$, with rational functions of the color as coefficients.

Proof. At $u = m$, the Gamma quotient in (6.7) is $\prod_{j=1}^{m-1}(1 + z/j)$, hence $Q_q(m) = 0$ for $q \geq m$. The limit in (6.8) is $(-1)^m Q_q'(m)\Phi(\omega, 1-m, a)$. Differentiating the quotient in u gives

$$Q_q'(m) = q![z^q] \prod_{j=1}^{m-1} (1 + z/j)(\psi(m+z) - \psi(m)).$$

The rational terms in the digamma recurrence become polynomials of degree at most $m-1$ after multiplication by the finite product. They do not contribute at degree q. Expanding $\psi(1+z) - \psi(1) = \sum_{r \geq 1} (-1)^{r+1} \zeta(r+1) z^r$ proves $(-1)^m Q_q'(m) = L_{m,q}$ in the finite form. Expanding the zeta values and using

$$\prod_{j=1}^{m-1} (1 - \ell/j) = (-1)^{m-1} \binom{\ell-1}{m-1}$$

gives (7.1). □

There is an independent elementary proof for $|\omega| < 1$. In that disk,

$$C_q^{\omega}(x,a) = \sum_{\ell=0}^{\infty} \frac{\omega^{\ell}}{x+\ell+a} \log^q \frac{x}{x+\ell+a}. \tag{7.4}$$

For $c > 0$, the substitution $t = x/(x+c)$ gives

$$\int_0^{\infty} \frac{x^{m-1}}{x+c} \log^q \frac{x}{x+c}\, \mathrm{d}x = c^{m-1} \int_0^1 \frac{t^{m-1} \log^q t}{(1-t)^m}\, \mathrm{d}t = c^{m-1} L_{m,q}. \tag{7.5}$$

Absolute summation proves the theorem in the disk, and locally normal continuation in ω gives the rest of Ω. At $\omega = -1$, one must not replace that continuation by the divergent series $\sum (-1)^{\ell}(\ell+a)^{m-1}$.

7.1 Low-order examples and logarithmic insertions

For every $q \geq 1$ the unweighted moment is independent of the shift:

$$\boxed{\int_0^{\infty} C_q^{\omega}(x,a)\, \mathrm{d}x = \frac{(-1)^q q! \zeta(q+1)}{1-\omega}.} \tag{7.6}$$

For example,

$$\int_0^{\infty} C_1^{\omega}(x,a)\, \mathrm{d}x = -\frac{\zeta(2)}{1-\omega}, \tag{7.7}$$

$$\int_0^{\infty} x C_2^{\omega}(x,a)\, \mathrm{d}x = 2(\zeta(2)-\zeta(3)) \left(\frac{a}{1-\omega} + \frac{\omega}{(1-\omega)^2} \right). \tag{7.8}$$

All fixed logarithmic insertions are equally explicit. For $q \geq m$ and $r \geq 0$,

$$\begin{aligned} &\int_0^{\infty} x^{m-1} \log^r x\, C_q^{\omega}(x,a)\, \mathrm{d}x \\ &\quad = (-1)^m r! [t^r] \left\{ \frac{\pi t}{\sin \pi t} \frac{Q_q(m+t)}{t} \Phi(\omega, 1-m-t, a) \right\}. \end{aligned} \tag{7.9}$$

The quotient $Q_q(m+t)/t$ is holomorphic. This is a finite coefficient formula in Lerch order derivatives, Gamma derivatives, and generalized harmonic polynomials. It follows by differentiating the normally convergent Mellin integral, not by termwise integration of its individual divergent logarithmic summands.

7.2 A precise generalized Stieltjes dictionary

Let $\omega^d = 1$ with $\omega \neq 1$. Introduce the incoming uncolored kernel [3]

$$C_q(x,a) = \frac{\log^{q+1} x}{q+1} + \sum_{j=0}^{q} (-1)^j \binom{q}{j} \gamma_j(x+a) \log^{q-j} x. \tag{7.10}$$

The finite residue-class identity for Lerch's function gives

$$\boxed{C_q^{\omega}(x,a) = \frac{1}{d} \sum_{r=0}^{d-1} \omega^r C_q\left(\frac{x}{d}, \frac{a+r}{d} \right).} \tag{7.11}$$

Indeed $\Phi(\omega, s, x+a) = d^{-s} \sum_r \omega^r \zeta(s, (x+a+r)/d)$; the $1/z$ poles and the $\log^{q+1}(x/d)$ terms cancel because $\sum_r \omega^r = 0$. This identity translates every preceding colored formula into an

ordinary integral of a finite combination of generalized Stieltjes constants. It also verifies the normalization against the incoming finite-filter calculus.

For the alternating color, define

$$a_- = \frac{x+a}{2}, \qquad a_+ = \frac{x+a+1}{2}.$$

Then, explicitly,

$$C_1^{-1}(x,a) = \frac{1}{2}\left[(\psi(a_+) - \psi(a_-))\log\frac{x}{2} + \gamma_1(a_+) - \gamma_1(a_-)\right]. \tag{7.12}$$

Its ordinary integral is $-\pi^2/12$ for every $a > 0$.

7.3 Exceptional moments and Dirichlet beta values

Theorem 7.2 (A zero of the leading colored tail)**.** *If $m \geq 1$ and $\Phi(\omega, 1-m, a) = 0$, then*

$$\boxed{\int_0^\infty x^{m-1} C_{m-1}^{\omega}(x,a)\,\mathrm{d}x = (-1)^{m+1}\partial_s\Phi(\omega, 1-m, a).} \tag{7.13}$$

The integral is ordinary and absolutely convergent.

Proof. The zero hypothesis removes the leading x^{-m} kernel term in Lemma 6.2, so the Mellin integral extends holomorphically across $u = m$. At $u = m$, $Q_{m-1}(m) = 1$ and $\Phi(\omega, 1-u, a) = -(u-m)\partial_s\Phi(\omega, 1-m, a) + O((u-m)^2)$. The removable limit in Theorem 6.5 is (7.13). □

Let $\beta(s) = \sum_{n\geq 0}(-1)^n(2n+1)^{-s}$ denote Dirichlet's beta function, continued in its order. Since $\Phi(-1, s, 1/2) = 2^s\beta(s)$, its negative odd zeros give the following family:

Corollary 7.3 (Alternating centered exceptional moments)**.** *For every integer $r \geq 1$,*

$$\boxed{\int_0^\infty x^{2r-1} C_{2r-1}^{-1}(x, \tfrac{1}{2})\,\mathrm{d}x = \frac{(-1)^r(2r-1)!}{\pi^{2r-1}}\beta(2r).} \tag{7.14}$$

In particular, writing $G = \beta(2)$ for Catalan's constant,

$$\int_0^\infty x C_1^{-1}(x, \tfrac{1}{2})\,\mathrm{d}x = -\frac{G}{\pi}. \tag{7.15}$$

Proof. The completed beta function $(4/\pi)^{(s+1)/2}\Gamma((s+1)/2)\beta(s)$ is invariant under $s \mapsto 1-s$ [13]. Comparing its finite values at $s = 1-2r$ and $s = 2r$ yields

$$\beta'(1-2r) = (-1)^{r-1}2^{2r-1}(2r-1)!\pi^{1-2r}\beta(2r).$$

Substitute this into Theorem 7.2. The classical functional equation is the input; the ordinary Stieltjes-difference integral is its consequence in the new kernel. □

7.4 Anchored shift primitives

For $a, a_0 > 0$ the quotient below is regular at $z = 0$, because $\Phi(\omega, 0, A) = 1/(1-\omega)$ does not depend on A:

$$\int_{a_0}^{a} C_q^{\omega}(x,v)\,\mathrm{d}v = -q![z^q]\,\frac{x^z}{z}\left(\Phi(\omega, z, x+a) - \Phi(\omega, z, x+a_0)\right). \tag{7.16}$$

This follows from $\partial_A\Phi(\omega, z, A) = -z\Phi(\omega, z+1, A)$. It gives every integration constant through the explicit anchor. At order zero and a root of unity, the Hurwitz identity $\zeta'(0, A) = \log\Gamma(A) - \frac{1}{2}\log(2\pi)$ gives the finite Gamma expression

$$\int_{a_0}^{a} C_0^{\omega}(x,v)\,\mathrm{d}v = -\sum_{r=0}^{d-1}\omega^r\log\frac{\Gamma((x+a+r)/d)}{\Gamma((x+a_0+r)/d)}. \tag{7.17}$$

All Gamma arguments are positive, and their logarithms are real before multiplication by the complex coefficients.

8 The color-one limit and its exact constant term

The formulas above are not uniform at $\omega = 1$. The precise normalization can nevertheless be determined, and it reconciles the colored moment with its uncolored Stieltjes counterpart. Fix $q \geq m \geq 1$, let $\omega = e^{-\lambda}$ with $\lambda > 0$, and set

$$M_{m,q}(\lambda, a) = \int_0^\infty x^{m-1} C_q^{e^{-\lambda}}(x, a)\, \mathrm{d}x.$$

Theorem 8.1 (Complete local color expansion)**.** *For $0 < \lambda < 2\pi$,*

$$\boxed{e^{-a\lambda} M_{m,q}(\lambda, a) = L_{m,q} \left\{ \frac{\Gamma(m)}{\lambda^m} + \sum_{k=0}^{\infty} \frac{(-\lambda)^k}{k!} \zeta(1 - m - k, a) \right\}.} \tag{8.1}$$

The series is convergent. In the cutoff coordinate λ,

$$\lim_{\lambda \downarrow 0} \left(e^{-a\lambda} M_{m,q} - L_{m,q} \Gamma(m) \lambda^{-m} \right) = -\frac{L_{m,q}}{m} B_m(a), \tag{8.2}$$

$$\mathrm{CT}_{\lambda=0}\, M_{m,q}(\lambda, a) = \frac{L_{m,q}}{m} (a^m - B_m(a)). \tag{8.3}$$

The first line is exactly the uncolored ordinary moment $\int_0^\infty x^{m-1} C_q(x, a)\, \mathrm{d}x$ from [3].

Proof. The classical convergent Lerch expansion [12, 25.14.3_1] is

$$e^{-a\lambda} \Phi(e^{-\lambda}, s, a) = \Gamma(1 - s) \lambda^{s-1} + \sum_{k=0}^{\infty} \zeta(s - k, a) \frac{(-\lambda)^k}{k!}.$$

Set $s = 1 - m$ and use Theorem 7.1. Its constant term is $L_{m,q} \zeta(1 - m, a) = -L_{m,q} B_m(a)/m$. Multiplying by $e^{a\lambda}$ adds the constant contribution $L_{m,q} \Gamma(m) a^m / m! = L_{m,q} a^m / m$ from the pole. This proves both normalizations. The incoming uncolored coefficient is $\kappa_{m,q} = -L_{m,q}/m$, so its moment is the first limit. □

For example, $M_{1,1} = -\zeta(2)/(1 - e^{-\lambda})$ is independent of a, and its raw constant term is $-\zeta(2)/2$. The normalized limit in (8.2), however, is $\zeta(2)(a - \frac{1}{2})$. Both formulas are correct in their stated normalizations. They cannot be identified by discarding the $e^{-a\lambda}$ factor.

All logarithmic moment versions follow by differentiating the local Mellin expression in u before putting $u = m$. Derivatives of λ^{-u} produce explicit powers of $\log \lambda$. Thus the scalar expansion already specifies the complete mechanism for those contacts. It does not assert that the pointwise limit of the colored kernels can be integrated across the entire half-line.

9 An explicit polylogarithm dictionary for two mass clusters

The incoming *Lattice Contacts and Mixed Zeta Identities* report evaluates a Gamma face when all but one mass agree. Its spectral-face section then asks for a finite dictionary for two distinct mass clusters; its research question Q4 asks more broadly about displaced masses and the dependence of depth on the number of clusters [4]. We evaluate the entire two-cluster sector, with arbitrary positive cluster multiplicities. The result is a finite combination of ordinary one-variable multiple polylogarithms inside an absolutely convergent spectral sum. In particular, two equal displaced masses require depth at most three, independently of the total number of factors.

The argument uses classical beta integration, aggregation of Dirichlet parameters, the Gamma product, and coefficient extraction. The transformation under exchange of clusters is a specialization of classical multiple-polylogarithm Landen formulas [7]; it is not asserted to be a new general transformation law. The contribution relative to the inspected incoming source is the explicit all-multiplicity dictionary and its convergence and depth statement. Minimality of the resulting depth and a bound depending only on the number of distinct clusters remain unproved.

9.1 The ordinary face and its independent regulators

Let $k \geq 2$, $b_1, \dots, b_k \in \mathbb{R}$, and

$$\Sigma_k = \{(t_1, \dots, t_k) : t_i \geq 0,\ \sum_i t_i = 1\}.$$

The measure dt means $dt_1 \cdots dt_{k-1}$ after eliminating t_k; it has no additional Euclidean surface factor. For nonempty $T \subseteq \{1, \dots, k\}$ set

$$t_T = \sum_{i \in T} t_i, \qquad b_T = \frac{\sum_{i \in T} t_i b_i}{t_T},$$

and give $t_T F(b_T)$ the value zero when $t_T = 0$. Define

$$\mathcal{H}_k(F; t) = \sum_{\varnothing \neq T \subseteq \{1,\dots,k\}} (-1)^{k-|T|} t_T F(b_T), \tag{9.1}$$

$$I_k(F; b_1, \dots, b_k) = \int_{\Sigma_k} \mathcal{H}_k(F; t) \frac{dt}{t_1 \cdots t_k}. \tag{9.2}$$

Constants and affine functions are annihilated.

These are ordinary integrals. To see convergence for $F \in C^1$ on the mass interval, each perspective is Lipschitz because

$$\partial_{t_i}\{t_T F(b_T)\} = F(b_T) + F'(b_T)(b_i - b_T), \qquad i \in T.$$

The alternating sum vanishes on every coordinate face, so $|\mathcal{H}_k(F; t)| \leq C \min_i t_i$. On the region where t_j is maximal, $t_j \geq 1/k$ and

$$\min_i t_i \leq \left(\prod_{i \neq j} t_i \right)^{1/(k-1)}.$$

Consequently the integral is absolutely convergent. The same argument allows sufficiently small independent complex powers $t_i^{\alpha_i}$ and any fixed powers of their logarithms.

For $\alpha_i > 0$, $A = \sum_i \alpha_i$, let

$$D_F(\boldsymbol{\alpha}; \boldsymbol{b}) = \frac{\Gamma(A)}{\prod_i \Gamma(\alpha_i)} \int_{\Sigma_k} F\left(\sum_i t_i b_i \right) \prod_i t_i^{\alpha_i - 1}\, dt.$$

If $A_T = \sum_{i \in T} \alpha_i$, the classical beta integral [9] gives

$$\frac{\Gamma(1 + A)}{\prod_i \Gamma(1 + \alpha_i)} \int_{\Sigma_k} \mathcal{H}_k(F; t) \prod_i t_i^{\alpha_i - 1}\, dt = \frac{1}{\prod_i \alpha_i} \sum_{\varnothing \neq T} (-1)^{k-|T|} A_T D_F(\boldsymbol{\alpha}_T; \boldsymbol{b}_T). \tag{9.3}$$

Here the notation on the right denotes the Dirichlet mean formed only from the labels in T, not the random barycenter on the left. The identity follows from $\mathbb{E}[t_T F(b_T)] = (A_T/A) D_F(\boldsymbol{\alpha}_T; \boldsymbol{b}_T)$. It is important that the α_i are independent until the squarefree coefficient is taken. The left side is holomorphic near zero and has value $I_k(F; \boldsymbol{b})$ there.

9.2 One scalar beta integral for arbitrary multiplicities

Let $r, s \geq 1$, $k = r + s$, and write B^r, C^s for r copies of B followed by s copies of C. Put $\delta = C - B$ and

$$E_F(t) = F(B + \delta t) - (1 - t) F(B) - t F(C), \tag{9.4}$$

$$Q(X, Y) = \frac{\Gamma(1 + X + Y)}{\Gamma(1 + X)\Gamma(1 + Y)}. \tag{9.5}$$

Theorem 9.1 (Scalar beta representation for two clusters)**.** *For $F \in C^2$ on the interval with endpoints B, C,*

$$\boxed{I_{r+s}(F; B^r, C^s) = r!s!\,[X^{r-1}Y^{s-1}]\,Q(X,Y)\int_0^1 \frac{E_F(t)}{t(1-t)}(1-t)^X t^Y\,dt.} \tag{9.6}$$

The integral on the right is holomorphic, for example for $\Re X, \Re Y > -1$, and the coefficient may be taken under the integral.

Proof. Aggregate the independent regulators belonging to the B cluster into X and those belonging to the C cluster into Y. The proportion of the C cluster has beta density $t^{Y-1}(1-t)^{X-1}/\mathrm{B}(Y,X)$. Thus the full-set term AD_F from (9.3) becomes

$$\begin{aligned}\mathcal{M}_F(X,Y) &= \frac{\Gamma(1+X+Y)}{\Gamma(X)\Gamma(Y)}\int_0^1 F(B+\delta t)t^{Y-1}(1-t)^{X-1}\,dt \\ &= XF(B) + YF(C) + XYQ(X,Y)\int_0^1 \frac{E_F(t)}{t(1-t)}(1-t)^X t^Y\,dt. \end{aligned} \tag{9.7}$$

The second line follows by subtracting the endpoint chord and integrating it exactly. Since $E_F(t) = O(t(1-t))$, it proves holomorphy of the full-set term near $(0,0)$ without leaving unremoved beta poles.

In the alternating sum in (9.3), each proper-subset term is holomorphic by the same chord subtraction, or is linear when it contains only one cluster. Every proper subset omits at least one regulator. Its coefficient of $\alpha_1 \cdots \alpha_k$ is therefore zero. In the full-set term, the corresponding coefficient is $r!s![X^rY^s]\mathcal{M}_F(X,Y)$: the squarefree coefficient in the rth power of a sum of r independent variables is $r!$. The chord term in (9.7) is linear and contributes nothing. Taking the removable value of (9.3) proves (9.6). Integrable powers at the two endpoints justify every fixed coefficient extraction. □

All kernels in this formula are explicit finite polynomials. Indeed,

$$\log Q(X,Y) = \sum_{n=2}^{\infty}\frac{(-1)^n\zeta(n)}{n}\big((X+Y)^n - X^n - Y^n\big). \tag{9.8}$$

For instance, with $L = \log(1-t)$ and $M = \log t$,

$$[XY]Qe^{XL+YM} = LM + \zeta(2), \tag{9.9}$$

$$[X^2Y]Qe^{XL+YM} = \tfrac{1}{2}L^2M + \zeta(2)L - \zeta(3), \tag{9.10}$$

$$[X^2Y^2]Qe^{XL+YM} = \tfrac{1}{4}L^2M^2 + \zeta(2)LM - \zeta(3)(L+M) + \tfrac{3}{2}\zeta(4) + \tfrac{1}{2}\zeta(2)^2. \tag{9.11}$$

These provide ordinary log-Gamma integral identities without any endpoint finite-part prescription.

9.3 The finite composition sum

Our one-variable multiple polylogarithm convention is

$$\mathrm{Li}_{a_0,a_1,\ldots,a_d}(z) = \sum_{n_0>n_1>\cdots>n_d\geq 1}\frac{z^{n_0}}{n_0^{a_0}n_1^{a_1}\cdots n_d^{a_d}}, \qquad |z|<1. \tag{9.12}$$

Only the outer index receives z. Its branch on $\mathbb{C}\setminus[1,\infty)$ is continued from zero by the usual iterated integrals; there is no branch singularity at zero. For $r,s \geq 1$, $k = r+s$, define the finite combination

$$\boxed{P_{r,s}(z) = \sum_{d=1}^{s}\binom{k-d-1}{r-1}\sum_{\substack{p_1+\cdots+p_d=k-1\\ p_i\geq 1}}\mathrm{Li}_{1,p_1,\ldots,p_d}(z).} \tag{9.13}$$

Every term has weight k and depth at most $s+1$.

Lemma 9.2 (Exact logarithmic face)**.** *If $x > 0$ and $B, C \geq 0$, set $q = (C - B)/(x^2 + C)$. Then*

$$I_{r+s}(U \mapsto -\log(x^2 + U); B^r, C^s) = (-1)^{r+s-1} r!s! P_{r,s}(q). \tag{9.14}$$

Proof. First suppose $C \geq B$, so $0 \leq q < 1$. The logarithm is

$$-\log(x^2 + C) - \log(1 - q(1 - t)).$$

The constant contributes only a linear term to $\mathcal{M}_F(X, Y)$. Applying the beta moment to the convergent logarithmic series gives

$$\mathcal{M}_F(X, Y) + (X + Y)\log(x^2 + C) = X \sum_{m=1}^{\infty} \frac{q^m}{m} \prod_{j=1}^{m-1} \frac{1 + X/j}{1 + (X + Y)/j}. \tag{9.15}$$

This series and its parameter derivatives converge normally for $|q| < 1$ and X, Y in a sufficiently small fixed neighborhood of zero. One may use the Gamma-ratio expression for the product to obtain a polynomial bound in m, uniformly on compact parameter sets; the geometric factor q^m then dominates every derivative.

For a finite product,

$$\prod_{j=1}^{m-1} \frac{1 + X/j}{1 + (X + Y)/j} = \prod_{j=1}^{m-1} \left(1 - \frac{Y}{j + X + Y}\right).$$

Choose d nonconstant factors and expand their denominators. The coefficient of $X^{r-1}Y^s$ is

$$(-1)^{k-1} \sum_{d=1}^{s} \binom{k - d - 1}{r - 1} \sum_{\substack{p_1 + \cdots + p_d = k-1 \\ p_i \geq 1}} \sum_{m > j_1 > \cdots > j_d \geq 1} \frac{1}{j_1^{p_1} \cdots j_d^{p_d}}. \tag{9.16}$$

For clarity, the d selected factors supply $(-Y)^d$. Their denominator expansions supply a total power $(X + Y)^{k-d-1}$ and sign $(-1)^{k-d-1}$. The remaining coefficient is precisely $\binom{k-d-1}{r-1}$. This proves the sign and every binomial coefficient in (9.16); no relation among zeta values has been used. Summing in m identifies exactly (9.13). The coefficient identity from Theorem 9.1 proves (9.14) for $0 \leq q < 1$.

Both sides are holomorphic in q on $\mathbb{C} \setminus [1, \infty)$. On the integral side this follows by subtracting the endpoint chord of $-\log(1 - q(1 - t))$; on each compact subset the subtracted numerator is uniformly $O(t(1 - t))$. The identity theorem therefore includes $q < 0$, which is the case $C < B$. In particular, the analytic continuation uses real logarithms and real multiple polylogarithms on the whole real interval $q < 1$. □

9.4 Gamma faces, convergence, and depth

For $a > 0$ and $U \geq 0$ define

$$G_a(U) = \log\Gamma(a + i\sqrt{U}) + \log\Gamma(a - i\sqrt{U}) - \log(2\pi), \tag{9.17}$$

where the two logarithms are continued from positive real arguments. Their sum is real. The classical Gamma product [8] gives

$$G_a(U) = 2\log\Gamma(a) - \log(2\pi) - \sum_{n=0}^{\infty} \log\left(1 + \frac{U}{(n + a)^2}\right). \tag{9.18}$$

Theorem 9.3 (Every two-cluster Gamma face)**.** *For $a > 0$, $B, C \geq 0$ and $r, s \geq 1$,*

$$\boxed{I_{r+s}(G_a; B^r, C^s) = (-1)^{r+s-1} r!s! \sum_{n=0}^{\infty} P_{r,s}\left(\frac{C-B}{(n+a)^2+C}\right).} \tag{9.19}$$

The spectral sum is ordinary and absolutely convergent, locally normally in the stated real parameters. The same identity has a dictionary of depth at most $\min(r, s) + 1$*, obtained if necessary by exchanging the two clusters and using the continued branch on the negative real axis. This is an upper bound on a supplied representation, not a claim of minimal functional depth.*

Proof. Apply the ordinary face functional termwise to (9.18). This is justified directly in the scalar beta formula: the endpoint-chord remainder of the nth logarithm is bounded in C^0 by a constant times $t(1-t)(C-B)^2(n+a)^{-4}$ on each compact parameter set. The fixed logarithmic beta kernels are integrable. Alternatively, the second derivatives of the log terms converge normally and the face is continuous in the C^2 norm. Constants disappear, and Lemma 9.2 gives the displayed result.

Each $P_{r,s}(z)$ is $O(z^2)$ at zero, since its first index is one and its depth is at least two. The arguments in the tail are $O(n^{-2})$. Hence the spectral summands are $O(n^{-4})$, uniformly on compact subsets. A finite number of initial arguments may lie to the left of -1 if $C < B$; their analytically continued values are regular. This proves absolute and local normal convergence without a small-displacement hypothesis.

The first expression has depth at most $s + 1$. The same face with the two clusters exchanged has depth at most $r + 1$. Taking the smaller establishes the asserted bound. □

The symmetry just used gives the precise transformation

$$P_{r,s}(z) = P_{s,r}\left(\frac{z}{z-1}\right), \qquad z \in \mathbb{C} \setminus [1, \infty). \tag{9.20}$$

It also follows from the classical composition form of the multiple-polylogarithm Landen connection formula [7]. Here it provides a check on branch conventions and a useful way to choose the smaller-depth expression.

The first collision coefficient checks all factorial normalizations:

$$I_{r+s}(G_a; B^r, (B+\delta)^s) = \frac{(-1)^{r+s-1}}{2} rs(r+s-2)!\,\delta^2 \sum_{n=0}^{\infty} \left((n+a)^2 + B\right)^{-2} + O(\delta^3). \tag{9.21}$$

This agrees with the predecessor's factorial-variance monomial formula. The full dictionary is global for $B, C \geq 0$, whereas that local collision expansion need not converge at the desired displacement.

9.5 Two equal displaced masses and a worked ordinary integral

For $s = 1$, (9.13) becomes $P_{r,1} = \mathrm{Li}_{1,r}$ and recovers precisely the incoming one-displaced theorem. For $s = 2$ it gives the new all-multiplicity specialization

$$\boxed{P_{r,2}(z) = r\,\mathrm{Li}_{1,r+1}(z) + \sum_{j=1}^{r} \mathrm{Li}_{1,j,r+1-j}(z).} \tag{9.22}$$

Thus r equal base masses and two equal displaced masses have depth at most three for every r. No increase in the base multiplicity increases this bound.

For example, set $E_a(t) = G_a(B+(C-B)t) - (1-t)G_a(B) - tG_a(C)$ and $q_n = (C-B)/((n+a)^2+C)$. The case $r=s=2$ gives the completely ordinary identity

$$\int_0^1 \frac{E_a(t)}{t(1-t)} \bigl(\log t \log(1-t) + \zeta(2)\bigr)\, dt \\ = -\sum_{n=0}^{\infty} \bigl(2\,\mathrm{Li}_{1,3}(q_n) + \mathrm{Li}_{1,1,2}(q_n) + \mathrm{Li}_{1,2,1}(q_n)\bigr)\,. \tag{9.23}$$

At $a = 2.3$, $B = 0.2$, $C = 0.9$, the associated face $I_4(G_a; B, B, C, C)$ is

$$-0.083625430091628765150371429019493368221777213899683 3\ldots.$$

This decimal is an independently checked diagnostic; the identity is proved above for the full parameter domain.

9.6 Mass derivatives and anchored polylogarithmic primitives

Delete the first index one in (9.13) and put

$$R_{r,s}(z) = \sum_{d=1}^{s} \binom{r+s-d-1}{r-1} \sum_{\substack{p_1+\cdots+p_d=r+s-1 \\ p_i \geq 1}} \mathrm{Li}_{p_1,\ldots,p_d}(z). \tag{9.24}$$

The defining strict series gives $P'_{r,s}(z) = R_{r,s}(z)/(1-z)$. It follows that

$$\partial_C I_{r+s}(G_a; B^r, C^s) = (-1)^{r+s-1} r!s! \sum_{n=0}^{\infty} \frac{R_{r,s}(q_n)}{(n+a)^2+C}, \tag{9.25}$$

$$\partial_B I_{r+s}(G_a; B^r, C^s) = -(-1)^{r+s-1} r!s! \sum_{n=0}^{\infty} \frac{R_{r,s}(q_n)}{(n+a)^2+B}. \tag{9.26}$$

These are total derivatives of the repeated cluster parameters: ∂_C moves all s copies of C together. They should not be confused with a derivative in just one independently labelled mass. The displayed expressions have depth at most s; exchanging clusters gives a representation with depth at most $\min(r,s)$.

Termwise differentiation is legitimate. For the tail, the arguments lie in one fixed small disk, where $P(z) = O(z^2)$ and $R(z) = O(z)$; local holomorphic extensions and Cauchy estimates give normal convergence of every fixed parameter derivative. The finitely many initial summands stay off their cuts. Consequently these equations also give higher polygamma identities after further differentiation. Since $I_{r+s}(G_a; B^{r+s}) = 0$, integration of (9.25) from $C = B$ to the desired endpoint is an anchored antiderivative with no undetermined constant.

9.7 Verification and the exact scope of the resolved question

The supplied script compares two independently organized exact coefficient calculations: the incoming monomial-face formula at $0^r, 1^s$, and (9.13) after the substitution $z = u/(1+u)$. It checks $r, s \in \{1,2,3,4\}$ and monomial degrees 2 through 15, together with the corresponding Landen coefficients, for a total of 448 exact rational assertions. Four numerical comparisons at 90 decimal digits use the ordinary beta integral on one side and strict harmonic sums with a Hurwitz-zeta spectral tail on the other. Their absolute residuals range from 9.8×10^{-77} to 3.3×10^{-75}. One case has $C - B > a^2 + B$, outside the predecessor's local collision-series disk. These are floating-point diagnostics, not certified interval enclosures or proof-assistant proofs.

The result resolves the request at the end of the incoming spectral-face section for an explicit dictionary when the mass set has exactly two distinct values, allowing every multiplicity. It

partially advances Q4 of that report. It does not evaluate two *unequal* displaced masses above a third base mass, does not prove that every two-cluster expression reduces to depth two, and does not establish a minimal depth theorem. A precise next target is the three-cluster case B^r, C, D with $C \neq D$, where an aggregation into three independent regulator sums should retain the same ordinary face normalization.

10 An exact depth-two example with two Gaussian colors

The Family 2 discussion in the manuscript [1] concerns the left-hand side below. We evaluate it in a smaller set of classical constants by an explicit exact certificate. Our strict convention is

$$\mathrm{Li}_{a,b}(u,v) = \sum_{n>m>0} \frac{u^n v^m}{n^a m^b}. \tag{10.1}$$

In particular, both colors in $\mathrm{Li}_{a,b}(i,i)$ are attached to summation indices: its numerator is i^{n+m}. This is different from the one-variable value $\mathrm{Li}_{a,b}(i,1)$. Let $G = \sum_{k\geq 0}(-1)^k/(2k+1)^2$ be Catalan's constant.

Theorem 10.1 (A weight-three two-color reduction)**.** *In the convention* (10.1)*,*

$$\boxed{\mathrm{Re}(\mathrm{Li}_{1,2}(i,i) - \mathrm{Li}_{2,1}(i,i)) = \frac{9}{32}\zeta(3) - \frac{\pi G}{4} + \frac{7\pi^2 \log 2}{96}.} \tag{10.2}$$

The proof below is a finite rational certificate using double-shuffle and distribution identities. It does not use numerical recognition or an unproved inversion formula.

10.1 Convergent relations, including the leading-one limit

All the series used below converge in their ordinary outer-index order. Indeed, every $\mathrm{Li}_{2,1}(u,v)$ with $|u| = |v| = 1$ converges absolutely, since $\sum_{n\geq 1} H_{n-1}/n^2 < \infty$. For $u \neq 1$, $|u| = |v| = 1$, write

$$\sum_{m<n} \frac{v^m}{m^2} = \mathrm{Li}_2(v) + O(n^{-1}).$$

Then $\mathrm{Li}_{1,2}(u,v)$ is the sum of $\mathrm{Li}_2(v)\sum_{n\geq 1} u^n/n$ and an absolutely convergent remainder. The single leading-one values below also have colors different from one.

For x, y among the fourth roots of unity, define

$$\begin{aligned}\mathcal{R}(x,y) = {} & \mathrm{Li}_{1,2}(x, y/x) + \mathrm{Li}_{2,1}(x, y/x) + \mathrm{Li}_{2,1}(y, x/y) \\ & - \mathrm{Li}_{1,2}(x,y) - \mathrm{Li}_{2,1}(y,x) - \mathrm{Li}_3(xy).\end{aligned} \tag{10.3}$$

When $x = 1$, the two identical leading-one terms in this expression are canceled *before* evaluation. Thus the definition is then

$$\mathcal{R}(1,y) = \mathrm{Li}_{2,1}(1,y) + \mathrm{Li}_{2,1}(y,y^{-1}) - \mathrm{Li}_{2,1}(y,1) - \mathrm{Li}_3(y). \tag{10.4}$$

Lemma 10.2. *Every value $\mathcal{R}(x,y)$ used here is zero, including $x = 1$.*

Proof. For $x \neq 1$, the strict stuffle product gives

$$\mathrm{Li}_1(x)\,\mathrm{Li}_2(y) = \mathrm{Li}_{1,2}(x,y) + \mathrm{Li}_{2,1}(y,x) + \mathrm{Li}_3(xy). \tag{10.5}$$

One may obtain it by multiplying finite sums with the same cutoff and partitioning the square into $n > m$, $n < m$, and $n = m$, then using the convergence just established. For the shuffle

product, use $\omega_0 = \mathrm{d}t/t$ and $\omega_c = c\,\mathrm{d}t/(1-ct)$ for $c \neq 0$. Define the endpoint-one iterated integral with c_1 outermost by

$$J(c_1, \dots, c_w) = \int_{0<t_w<\cdots<t_1<1} \prod_{j=1}^{w} \omega_{c_j}(t_j).$$

Expanding the kernels below endpoint one and then taking the convergent endpoint limit gives

$$\mathrm{Li}_{1,2}(u,v) = J(u,0,uv), \qquad \mathrm{Li}_{2,1}(u,v) = J(0,u,uv).$$

The three interleavings of $J(x)$ and $J(0,y)$ therefore give

$$\mathrm{Li}_1(x)\,\mathrm{Li}_2(y) = \mathrm{Li}_{1,2}(x, y/x) + \mathrm{Li}_{2,1}(x, y/x) + \mathrm{Li}_{2,1}(y, x/y). \tag{10.6}$$

Subtracting (10.5) proves the claim for $x \neq 1$.

Here is a direct justification of the limiting relation at $x = 1$; no divergent series is assigned a value. Apply the two product formulas with $x = r \in (0,1)$ and keep $|y| = 1$ fixed. The difference of their leading-one terms satisfies

$$\begin{aligned}
&\mathrm{Li}_{1,2}(r, y/r) - \mathrm{Li}_{1,2}(r,y) \\
&\quad = \sum_{m,k\geq 1} \frac{y^m r^k (1-r^m)}{(m+k)m^2}, \\
&|\mathrm{Li}_{1,2}(r, y/r) - \mathrm{Li}_{1,2}(r,y)| \leq -\log(1-r)(\zeta(2) - \mathrm{Li}_2(r)) = O\Big((1-r)\log^2(1-r)\Big).
\end{aligned}$$

The bound tends to zero. For example, $\zeta(2) - \mathrm{Li}_2(r) = \int_r^1 -\log(1-t)\,\mathrm{d}t/t \leq (1-r)(1-\log(1-r))/r$. Each remaining depth-two term is dominated by $\sum_{n>m} n^{-2}m^{-1}$, and the trilogarithm converges absolutely. Letting $r \uparrow 1$ proves (10.4). In particular $\mathcal{R}(1,1) = 0$ is Euler's identity $\mathrm{Li}_{2,1}(1,1) = \zeta(3)$, with its proof included in this limit argument. □

For $u, v \in \{1, -1\}$ choose $\sqrt{1} = 1$ and $\sqrt{-1} = i$, and put

$$\mathcal{D}_s(u) := \mathrm{Li}_s(u) - 2^{s-1}(\mathrm{Li}_s(\sqrt{u}) + \mathrm{Li}_s(-\sqrt{u})), \tag{10.7}$$

$$\mathcal{D}_{a,b}(u,v) := \mathrm{Li}_{a,b}(u,v) - 2^{a+b-2} \sum_{\epsilon,\eta\in\{1,-1\}} \mathrm{Li}_{a,b}(\epsilon\sqrt{u}, \eta\sqrt{v}). \tag{10.8}$$

Only convergent instances occur in the table below: $s = 3$ with $u = \pm 1$, $s = 2$ with $u = 1$, $s = 1$ with $u = -1$, $(a,b) = (2,1)$ with the listed colors, and $(a,b) = (1,2)$ with $u = -1$. All these $\mathcal{D}$ values are zero. To prove this directly, sum the signs in a finite outer-cutoff sum. They select the even summation indices; replacing each index by twice a new index gives the factor 2^{d-w} at depth d and weight w. The outer cutoff becomes its integer half. Pass to the limit using the preceding convergence checks.

10.2 The complete rational certificate

Write $\delta_s = \mathrm{Li}_s(i) - \mathrm{Li}_s(-i)$ for $s = 1, 2$ and define

$$\mathcal{T} = 96(\mathrm{Li}_{1,2}(i,i) - \mathrm{Li}_{2,1}(i,i)) - 27\,\mathrm{Li}_3(1) - 24\delta_1\delta_2 + 42\,\mathrm{Li}_2(1)\,\mathrm{Li}_1(-1). \tag{10.9}$$

The following table specifies a finite formal identity:

$$\mathrm{Re}\,\mathcal{T} = \mathrm{Re}\sum_{j=1}^{15} c_j \mathcal{E}_j. \tag{10.10}$$

j	c_j	$\mathcal{E}_j$
1	-15	$\mathcal{D}_3(1)$
2	6	$\mathcal{D}_3(-1)$
3	-24	$\mathcal{R}(1,-1)$
4	60	$\mathcal{D}_{2,1}(-1,-1)$
5	96	$\mathcal{R}(i,1)$
6	192	$\mathcal{R}(i,-1)$
7	12	$\mathcal{D}_{2,1}(1,1)$
8	12	$\mathcal{R}(1,1)$
9	-168	$\mathrm{Li}_1(i)\mathcal{D}_2(1)$
10	90	$\mathcal{D}_{2,1}(1,-1)$
11	96	$\mathcal{R}(i,i)$
12	192	$\mathcal{R}(1,i)$
13	12	$\mathcal{D}_{1,2}(-1,-1)$
14	42	$\mathcal{D}_{1,2}(-1,1)$
15	-12	$\mathrm{Li}_2(-1)\mathcal{D}_1(-1)$

For a completely explicit check of (10.10), expand the definitions of $\mathcal{R}$ and $\mathcal{D}$, and expand the products by (10.5). Identify the real part of each coordinate with the real part of its complex conjugate. With

$$X_{a,b}^{p,q} = \operatorname{Re}\mathrm{Li}_{a,b}(i^p, i^q), \qquad Z_p = \operatorname{Re}\mathrm{Li}_3(i^p),$$

both sides of (10.10) become precisely

$$48X_{1,2}^{1,1} + 48X_{1,2}^{1,3} + 42X_{1,2}^{2,0} + 42X_{2,1}^{0,2} - 144X_{2,1}^{1,1} + 48X_{2,1}^{1,3} + 21Z_0 - 6Z_2. \tag{10.11}$$

This is an identity in a finite rational vector space before evaluating any period. The companion script independently constructs the six-term shuffle residuals directly in strict index-and-color coordinates, expands every row, and checks equality with exact fractions.

Proof of Theorem 10.1. Every $\mathcal{E}_j$ is zero by Lemma 10.2 or the finite-cutoff distribution identities, so (10.10) gives $\operatorname{Re}\mathcal{T} = 0$. The elementary single-value evaluations needed to simplify it are

$$\mathrm{Li}_1(i) = -\frac{\log 2}{2} + \frac{\pi i}{4}, \qquad \delta_1 = \frac{\pi i}{2}, \qquad \delta_2 = 2iG, \qquad \mathrm{Li}_2(1) = \frac{\pi^2}{6}, \qquad \mathrm{Li}_1(-1) = -\log 2.$$

Here the first identity follows from $\mathrm{Li}_1(z) = -\log(1-z)$ with the principal logarithm, and the formula for δ_2 follows by taking the odd terms of the absolutely convergent dilogarithm series. Substitution into (10.9) yields

$$0 = 96\operatorname{Re}(\mathrm{Li}_{1,2}(i,i) - \mathrm{Li}_{2,1}(i,i)) - 27\zeta(3) + 24\pi G - 7\pi^2\log 2.$$

Division by 96 proves (10.2). □

The ordinary stuffle product supplies, as a further consequence, the individual real values

$$\operatorname{Re}\mathrm{Li}_{2,1}(i,i) = \frac{15}{64}\zeta(3) - \frac{\pi^2\log 2}{32}, \tag{10.12}$$

$$\operatorname{Re}\mathrm{Li}_{1,2}(i,i) = \frac{33}{64}\zeta(3) - \frac{\pi G}{4} + \frac{\pi^2\log 2}{24}. \tag{10.13}$$

Indeed, $\mathrm{Li}_1(i)\,\mathrm{Li}_2(i) = \mathrm{Li}_{1,2}(i,i) + \mathrm{Li}_{2,1}(i,i) + \mathrm{Li}_3(-1)$, while $\operatorname{Re}\mathrm{Li}_2(i) = -\pi^2/48$ and $\mathrm{Li}_3(-1) = -3\zeta(3)/4$ by even–odd splitting. Together with (10.2), these identities determine the sum and difference of the two real values.

10.3 Reconciliation with the printed Family 2 expression

The manuscript's displayed Family 2 expression (`double:eq:fam2-example`) uses

$$L = \log 2, \qquad r = \frac{1+i}{2}, \qquad \mu_2 + i\lambda_2 = \mathrm{Li}_2(r), \qquad h_{a,b} = \mathrm{Re}\,\mathrm{Li}_{a,b}(i,1).$$

To prove that its right side agrees with (10.2), we first evaluate the required $h_{1,2}$ independently of the printed equality.

Lemma 10.3. *In the strict convention* (10.1),

$$h_{1,2} = -\frac{29}{32}\zeta(3) + \frac{\pi G}{4} + \frac{\pi^2 L}{96}. \tag{10.14}$$

Proof. Use the same δ_s and relation rows $\mathcal{E}_j$ as in the certificate (10.10), and put

$$\mathcal{T}_h = 96\,\mathrm{Li}_{1,2}(i,1) + 87\,\mathrm{Li}_3(1) + 24\delta_1\delta_2 + 6\,\mathrm{Li}_2(1)\,\mathrm{Li}_1(-1).$$

A second exact rational identity in those same coordinates is

$$\begin{aligned}\mathrm{Re}\,\mathcal{T}_h = \mathrm{Re}(&-9\mathcal{E}_1 + 18\mathcal{E}_2 - 12\mathcal{E}_4 - 96\mathcal{E}_5 - 24\mathcal{E}_9 + 6\mathcal{E}_{10}\\ &- 12\mathcal{E}_{13} + 6\mathcal{E}_{14} + 12\mathcal{E}_{15}) = 0.\end{aligned} \tag{10.15}$$

As before, this is checked just by expanding the displayed definitions, stuffle products, and conjugate pairs. Every row on the right vanishes by the already proved convergent identities; the only shuffle row needed here has first factor $\mathrm{Li}_1(i)$. Since $\delta_1\delta_2 = -\pi G$ and $\mathrm{Li}_2(1)\,\mathrm{Li}_1(-1) = -\pi^2 L/6$, the left side is $96h_{1,2} + 87\zeta(3) - 24\pi G - \pi^2 L$. Solving proves (10.14). The companion bridge verifier reconstructs these nine rows and checks the exact residual separately from the two-color identity. □

The remaining depth-one and logarithmic quantities are

$$\mu_2 = \frac{5\pi^2}{96} - \frac{L^2}{8}, \qquad \lambda_2 = G - \frac{\pi L}{8}, \qquad h_{1,1} = \frac{L^2}{8} - \frac{\pi^2}{32}. \tag{10.16}$$

For completeness, the classical dilogarithm reflection and Landen identities [7] give

$$2\mu_2 = \zeta(2) - \log r \log \bar{r}, \qquad \mathrm{Li}_2(i) + \mathrm{Li}_2(\bar{r}) = -\tfrac{1}{2}\log^2(1-i).$$

Use $\log r = -L/2 + i\pi/4$ and $\log(1-i) = L/2 - i\pi/4$ to obtain the first two evaluations in (10.16). These standard dilogarithm identities follow by differentiating and fixing the constant at zero, with continuation in the principal domains containing the displayed arguments. Finally, $\mathrm{Li}_{1,1}(i,1) = \frac{1}{2}\log^2(1-i)$ follows from the shuffle of two identical one-letter integrals, and gives the third evaluation.

Corollary 10.4 (The printed Family 2 equality)**.** *The displayed depth-two identity in the manuscript is valid:*

$$\mathrm{Re}(\mathrm{Li}_{1,2}(i,i) - \mathrm{Li}_{2,1}(i,i)) = 3\zeta(3) - 4L\mu_2 - \pi\lambda_2 - 4h_{1,1}L + 3h_{1,2}. \tag{10.17}$$

Proof. Substituting (10.14) and (10.16) into the right side gives

$$\begin{aligned}&\left(3 - \frac{87}{32}\right)\zeta(3) + \left(-1 + \frac{3}{4}\right)\pi G + \left(-\frac{20}{96} + \frac{12}{96} + \frac{12}{96} + \frac{3}{96}\right)\pi^2 L + \left(\frac{1}{2} - \frac{1}{2}\right)L^3\\ &\quad = \frac{9}{32}\zeta(3) - \frac{\pi G}{4} + \frac{7\pi^2 L}{96}.\end{aligned}$$

This equals the independently proved left side by Theorem 10.1. □

Independent diagnostic. For verification against the strict convention, direct differentiation and integration of (10.1) give

$$\operatorname{Re}\operatorname{Li}_{1,2}(i,i)=\operatorname{Re}\int_0^1\frac{i\operatorname{Li}_2(-t)}{1-it}\,\mathrm{d}t,\qquad \operatorname{Re}\operatorname{Li}_{2,1}(i,i)=-\int_0^1\frac{t\log t\log(1+t)}{1+t^2}\,\mathrm{d}t.$$

An optional 80-digit quadrature in the companion script gives

$$\operatorname{Re}(\operatorname{Li}_{1,2}(i,i)-\operatorname{Li}_{2,1}(i,i))=0.11751017576784368621941000460386119840621435030463\ldots.$$

This is a numerical diagnostic; the finite relation and analytic arguments above prove the identity. No conclusion about S_6 or the revised S_8 conjecture follows from this example.

11 Audit, corrections, and integration status

11.1 What was actually checked

The source anchor is the repository revision `c4036d60715916ae72dd18983782fdb182840b86`. All eleven archives in the supplied incoming directory at that revision were unpacked and indexed. The review concentrated on their theorem and question ledgers and on the passages used by the present arguments. The manuscript review concentrated on Gaussian depth reduction, the S_6 obstruction, the revised S_8 candidate, and the harmonic/Gamma interfaces. This targeted review is not a certification of all manuscript volumes or every theorem in the incoming reports.

Three source questions have precise new outcomes:

1. **Ordered boundary report, research question 2.** The positive constants in `sections/08_gamma_basis.tex` are evaluated by Theorem 3.1. Its proof uses the source's established top-row identities and a classical integral with an explicitly checked normalization. The matrix T_m and all column orders are stated before use. The later-index determinant in (3.11) is a different object.

2. **Balanced Stieltjes report, research question 4.** The homogeneous colored balancing, its natural Lerch coordinates, its ordinary Mellin moments, and its finite root-of-unity filter are supplied in Sections 6–8. Uniqueness is asserted for coefficients independent of the shift and for a decay condition across an open interval of shifts. This is the same kind of hypothesis needed in the predecessor's uncolored uniqueness theorem.

3. **Lattice-contact report, end of the spectral-face section and research question Q4.** Theorem 9.3 treats exactly two distinct mass values at every pair of positive multiplicities. It supplies a finite dictionary per spectral term. A problem involving two unequal displaced masses above a third base value has three clusters and remains outside this theorem.

The companion source manifest records the archive names and their SHA-256 hashes. The integration notes list exact predecessor paths and theorem labels, so a later integration can reconcile the new results with a changed repository head.

11.2 Corrections and qualifications

No false mathematical theorem was identified in the particular incoming passages used for these three developments. It would be misleading to manufacture an erratum from an open question that the source already labels correctly. The following distinctions are nevertheless essential when integrating the continuation.

The colored generator changes. For a nontrivial color, $\Phi(\omega, s, a)$ has no pole at $s = 1$. The appropriate balanced kernel is (6.3). Copying the extra uncolored term $\log^{q+1} x/(q+1)$ would spoil its large-x cancellation. This is a correction to that potential extension, not a claim that the incoming uncolored definition is erroneous.

Fixed-shift uniqueness can fail. The open-interval clause in Theorem 6.3 cannot simply be deleted on the full color domain. The exceptional-color example following that theorem exhibits simultaneous vanishing of consecutive tail coordinates. At such a point, a logarithm times a lower balanced kernel can preserve the required decay.

Constant terms depend on the chosen expression. The raw λ-constant term of $M_{m,q}(\lambda, a)$ differs from that of $e^{-a\lambda}M_{m,q}(\lambda, a)$ by $L_{m,q}a^m/m$. Equations (8.2) and (8.3) retain that term explicitly. This avoids an incorrect identification of a color limit with an uncolored ordinary moment.

Positivity does not settle every later determinant. The positive density (4.5) evaluates the normalizing determinant. The source's later rows involve $\Im(x + i/2)^{2r}$ with oscillatory signs. One cannot infer all later signs by applying positivity of a Vandermonde product to a different matrix. In particular, no all-pairs nonvanishing claim follows from the closed formula for K_m alone.

Finite normal moments do not give a finite-dimensional normal distribution. The statistic in Theorem 5.1 is bounded at fixed n, σ. We prove equality of moments only through degree $n + 1$, and we do not assert that this range is maximal. The distributional limit in Corollary 5.3 has a separate moment-method proof.

Branch and color conventions remain explicit. The two-cluster $\mathrm{Li}_{\boldsymbol{a}}(z)$ assigns its argument only to the largest summation index. Colored depth-two notation uses two independent color factors and is defined separately. When $C < B$, the cluster formula uses the continuation on $\mathbb{C} \setminus [1, \infty)$; substituting a divergent power series at a negative argument of modulus greater than one is not valid. Likewise, the ordinary ω series for the colored balanced kernel is used only inside its disk of convergence before continuation to the full slit domain.

A small editorial correction. At the pinned revision, the manuscript README repeats consecutively the paragraph describing the complete 1,032-input first-letter X family, its rank 1,008, and its kernel dimension 24. The optional patch in the integration directory removes one copy and leaves the mathematical statements unchanged.

11.3 Evidence and its limits

Analytic proofs establish the identities. The accompanying exact and numerical calculations provide independent checks on the algebra, normalization, and selected parameter values.

The Gamma normalization suite evaluates the original coefficient matrices at $m = 1, 2, 3$, the parameterized determinants at unequal real and complex shifts, and scalar digamma/polygamma insertions. It contains fourteen checks at 42 decimal digits; the largest recorded relative residual is approximately 7.1×10^{-43}. The finite-Gaussian suite constructs exponential jets of ordinary one-dimensional moment integrals and takes their determinant by finite polynomial arithmetic. It does not use the paired Gamma product on the left side. Its fifteen checks include the fourth moment at $n = 3$ and non-quarter shifts; the largest recorded absolute residual is approximately 4.9×10^{-41}.

The cluster suite contains 448 exact rational assertions, comparing two different coefficient constructions and checking cluster exchange. Four independent numerical comparisons at 90 digits include a displacement outside the predecessor's local collision-series disk. The colored suite supplies 41 exact assertions and eleven numerical checks at 50 digits, covering finite tail and zeta coefficients, a simultaneous-tail-zero example, ordinary moments, root-of-unity specializations, and an anchored primitive. Its largest numerical residual is approximately 5.4×10^{-37}. Three further 55-digit checks evaluate a negative cluster argument $q = -4$, its Landen transform, and the oriented anchored primitive. The separate depth-two certificate uses exact rational relations, with its conventions and limiting argument explained in the corresponding section. A second certificate, using nine of the same relation rows, evaluates the intermediate $h_{1,2}$ and reconciles the actual printed Family 2 right side exactly.

The JSON records specify the checks actually performed. Floating point residuals are diagnostics, not certified error intervals. An exact certificate verifies an algebraic reduction under its stated analytic identities; it is not by itself a formalization of convergence in a proof assistant. No decimal relation search is promoted to a proof of S_6 or S_8.

12 Further research questions

The results suggest several concrete next problems. The order below prioritizes exact identities and structural classification over estimates. Where a question is already in the supplied sources, the proposed next step is stated relative to the new formulas.

Q1. Find the first non-Gaussian moment and its exact defect. For the statistic $X_{n,\sigma}$, determine

$$\mathbb{E}X_{n,\sigma}^{n+2} - \mathbb{E}Z^{n+2}, \qquad Z \sim N(0,1),$$

and decide whether $n+1$ is the exact universal matching degree. The distinct-parameter proof stops because there are only $n+1$ pair labels. Repeated powers of one label bring in $S_3, S_5, \ldots$; the complete generating identity (4.8) gives coupled equations for these new coordinates. This is an algebraic obstruction to isolate, not evidence that the next moment must be nonzero in every case.

Q2. Classify joint polygamma moments. Give a finite combinatorial rule for the mixed cumulants of $S_1, S_3, S_5, \ldots$ obtainable from the paired deformation. The logarithm of the Gamma product involves interactions of at most two parameter labels, but repeated powers of a label carry higher polygamma statistics. A useful theorem would state precisely which joint moments can be separated at dimension n, and give their coefficients as polygamma values at 2σ. At $\sigma = 1/4$, reflection should yield especially short rational combinations of powers of π.

Q3. Extend the determinant evaluation to later source rows. For the determinant in (3.11), express a prescribed family of later row sets as the evaluated Gustafson normalizer times an explicit symmetric polynomial or a finite hypergeometric sum. Such an identity would preserve the oscillatory polynomial insertions exactly. It could address the source's fixed-gap sign changes without an unjustified total-positivity argument. The gap-six statement $\det F_{b,3} > 0$ for $b \geq 7$ remains a concrete open target [2].

Q4. Classify simultaneous zeros of colored tails. For $A_k(\omega, a) = (\omega\partial_\omega + a)^k(1-\omega)^{-1}$, classify all common zeros of consecutive A_k at $a > 0$ and $\omega \in \Omega$. The example in this article proves that consecutive zeros do occur. Determine the resulting dimension of the space of homogeneous combinations with enhanced decay, and the exact extra ordinary moments that each zero pattern permits. A classification on the unit circle may be simpler than one on the entire slit domain.

Q5. Give a character-theoretic exceptional-moment dictionary. Apply finite combinations of root-of-unity colors to isolate Dirichlet characters. At a negative integer where the relevant L-function vanishes, determine the corresponding ordinary Stieltjes-difference integral and simplify it using the functional equation. The alternating beta family (7.14) is the first case. A complete statement should distinguish primitive and imprimitive characters and retain their conductor and parity factors.

Q6. Evaluate the first bilinear colored moment. For positive shifts, determine a natural ordinary domain and an exact function class for

$$\int_0^\infty x^{u-1} C_p^\omega(x,a) C_q^\eta(x,b)\,\mathrm{d}x.$$

The scalar Mellin formula suggests a Mellin convolution, but the transform of a product is not the product of transforms. One target is a two-parameter beta/Feynman integral whose integer contacts reduce to a finite family of multiple-Lerch coordinates. Coincident shifts and inverse colors are plausible specializations to analyze first.

Q7. Resolve the three-cluster spectral face. For B^r, C, D with three distinct nonnegative masses, aggregate the independent regulators into three sums before extracting the squarefree coefficient. The two-cluster proof then suggests a two-simplex hypergeometric integral and a finite iterated integral dictionary. Determine the correct multivariable polylogarithm arguments and branch domain. The answer should explain what changes when $C = D$, where the depth-three formula (9.22) must be recovered.

Q8. Determine genuine depth reductions and lattice closures. The bound $\min(r,s)+1$ is a supplied representation bound. Can the two-cluster combinations be reduced further as functions? Independently, characterize parameters for which the outer sum in (9.19) collapses to finitely many multiple Gamma, Hurwitz, or polylogarithm coordinates. These are two different problems: reducing depth termwise does not eliminate an infinite spectral sum.

Q9. Iterate anchored primitives with their constants intact. The colored primitive (7.16) lowers the Lerch order by one; the cluster derivative deletes an initial index one in the polylogarithm word. Iterate both operations and identify a canonical Barnes-function normalization for the root-of-unity cases. At every stage, retain explicit anchors. Determine whether the normalized color-one limit commutes with these primitives and identify any additional polynomial terms when it does not.

Q10. Supply a new relation for S_6 and revised S_8. The manuscript already distinguishes the conjectural period equalities from its exact bounded-span and shuffle-obstruction theorems. A proof must introduce a relation outside the already excluded span, or a new analytic integral identity, with all branch and endpoint terms included. The weight-three worked reduction in this article is a useful small exact certificate; it does not remove the higher-weight obstruction. An appropriate deliverable is a rational certificate whose primitive rows have independent analytic proofs, not only an enlarged numerical relation search.

Q11. Formalize the reusable analytic interfaces. The most useful first targets are continuous Cauchy–Binet with dominated parameter differentiation, finite coefficient extraction from normally convergent Mellin germs, and the independent-regulator face identity (9.3). Formalizing these interfaces would allow many subsequent identities to be checked by finite algebra while retaining the convergence arguments that justify that algebra.

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[1] V. Reshetnikov, *ProveIt: Analysis/Polylogarithms manuscript*, repository revision `c4036d60715916ae72dd18983782fdb182840b86`, especially the Gaussian depth-reduction and harmonic/Gamma chapters. https://github.com/VladimirReshetnikov/ProveIt/tree/c4036d60715916ae72dd18983782fdb182840b86/Analysis/Polylogarithms/docs/manuscript.

[2] ProveIt incoming report, *Ordered Boundary Identities and Gamma Kernels*, 11 October 2026, archive `ProveIt_Ordered_Boundary_Identities_2026-10-11.zip`, inner `sections/08_gamma_basis.tex` (`gb:basis`, `gb:moment-identities`, `gb:normalization`) and `sections/09_audit_research.tex`, question 2, at revision `c4036d60715916ae72dd18983782fdb182840b86`. https://github.com/VladimirReshetnikov/ProveIt/tree/c4036d60715916ae72dd18983782fdb182840b86/docs/incoming.

[3] ProveIt incoming report, *Balanced Stieltjes Moments: Bernoulli Closure, Harmonic Filters, and Polylogarithmic Resonances*, 11 October 2026, archive `ProveIt_Balanced_Stieltjes_Moments_2026-10-11.zip`, inner `article/Balanced_Stieltjes_Moments.tex`, especially `bsm:thm:unique`, `bsm:thm:master`, `bsm:thm:integer`, and research question 4, at revision `c4036d60715916ae72dd18983782fdb182840b86`. Available in the incoming directory cited above.

[4] ProveIt incoming report, *Lattice Contacts and Mixed Zeta Identities*, 11 October 2026, archive `ProveIt_Lattice_Contacts_and_Mixed_Zeta_Identities_2026-10-11.zip`, inner `sections/07_spectral_faces.tex` (`sfj:thm:monomial`, `sfj:thm:gamma-polylog`) and `sections/08_audit_and_research.tex`, Q4, at revision `c4036d60715916ae72dd18983782fdb182840b86`. Available in the incoming directory cited above.

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[6] S. É. Derkachov and A. N. Manashov, *Spin Chains and Gustafson's Integrals*, Journal of Physics A: Mathematical and Theoretical **50** (2017), 294006; arXiv:1611.09593v2, equation (2). https://doi.org/10.1088/1751-8121/aa749a. https://arxiv.org/html/1611.09593v2.

[7] J.-I. Okuda and K. Ueno, *The Sum Formula of Multiple Zeta Values and Connection Problem of the Formal Knizhnik–Zamolodchikov Equation*, arXiv:math/0310259v2 (2004), especially equation (2). https://arxiv.org/abs/math/0310259.

[8] NIST Digital Library of Mathematical Functions, §5.8, *Infinite Products*, especially 5.8.3. https://dlmf.nist.gov/5.8.

[9] NIST Digital Library of Mathematical Functions, §5.12, *Beta Function*, especially Euler's beta integral. https://dlmf.nist.gov/5.12.

[10] NIST Digital Library of Mathematical Functions, §5.15, *Polygamma Functions*. https://dlmf.nist.gov/5.15.

[11] NIST Digital Library of Mathematical Functions, §25.11, *Hurwitz Zeta Function*. https://dlmf.nist.gov/25.11.

[12] NIST Digital Library of Mathematical Functions, §25.14, *Lerch's Transcendent*, especially 25.14.5 and 25.14.3_1. https://dlmf.nist.gov/25.14.

[13] NIST Digital Library of Mathematical Functions, §25.15, *Dirichlet L-Functions*, functional equation for primitive characters and trivial zeros. https://dlmf.nist.gov/25.15.